

Assignment 4: Literature Review & Analytical Questions (Lee, 2018)

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Task: Literature survey, questions, and discussion on "Exoplanets: past, present, and future" (Chien-Hsiu Lee, 2018)

Discord Submission Status Note:

"Please note: Due to platform restrictions and access limitations in my region, I am unable to access the course Discord server. As discussed and approved, I am submitting my formulated research questions and analytical notes regarding the weekly paper directly through this document in our shared Google Drive directory and my GitHub repository."

? Analytical Questions Derived from Lee (2018)

Question 1: Resolving Blending Effects and Radius Inflation in High-Cadence Surveys

In Section 3 (New Frontiers), the paper highlights a critical data-pipeline challenge: many transit surveys sacrifice spatial resolution for a larger field of view (FOV), leading to "blending effects" where unrelated background stars/galaxies bleed into the same CCD pixels. This causes an underestimation of the planet's true radius. *My question is:* How can automated robotic adaptive optics systems (like Robo-AO) be integrated into modern real-time automated data pipelines to flag and mathematically "deblend" light curves during the initial transit detection phase, rather than relying on years of slow follow-up observations?

Question 2: Resolving the Mass-Radius Composition Degeneracy for Volatile-Rich Worlds

In Section 2 (Status Quo), the review notes that while precise mass (from RV) and radius (from transit) estimates give us mean densities, a degenerated physical picture still remains: we cannot easily distinguish between a planet with a small rocky core and a large, volatile-rich (H/He or water) envelope, and one with a large rocky core and a very thin envelope. *My question is:* Looking toward the next generation of infrared space telescopes (like ARIEL, scheduled for 2028), how will transmission spectroscopy break this specific degeneracy by probing atmospheric chemical abundances, and what are the limitations when dealing with heavily cloudy/hazy atmospheres in hot Jupiters?

Question 3: Contradicting Mass Functions of Free-Floating Planets (MOA vs. OGLE-IV)

Section 2 discusses the observational conflict regarding the population of Free-Floating Planets (FFPs). The MOA team (Sumi et al., 2011) claimed an excess of Jovian-mass FFPs (suggesting they are twice as populous as main-sequence stars, likely formed in disks and ejected). However, the OGLE-IV survey (Mróz et al., 2017) refuted this excess, instead detecting shorter timescale events pointing toward terrestrial or super-Earth mass FFPs. *My question is:* If core-accretion and planet-planet scattering theories favor the ejection of lighter, terrestrial-mass planets rather than massive Jovian ones, what observational biases in early microlensing surveys might have over-represented the giant FFP population, and how will upcoming space-based microlensing missions (like WFIRST/Roman Space Telescope) settle this debate?